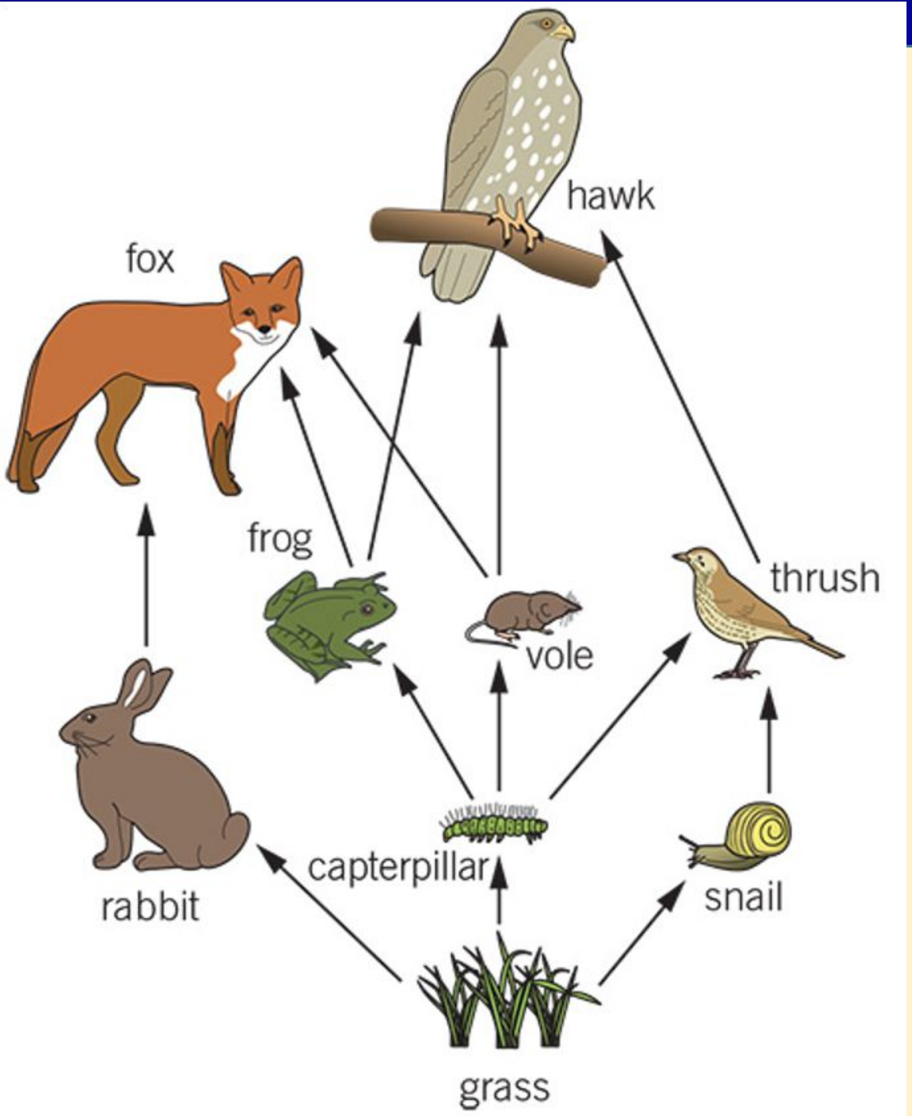
Title: Bioaccumulation



1. How many secondary consumers in this food web? **4**
2. What do you think would happen if all of the snails died? **The thrush would have less to eat/ would eat more caterpillars/ would decrease in numbers > hawk would have less to eat**
3. What is a decomposer?
   Organisms that break down decaying or dead organisms eg. some fungi, some invertebrates
4. What type of substances can decomposers break down? **organic / biodegradable materials**

Success criteria

Lesson Title	Working towards	Working at	Working beyond
What is an Ecosystem	I can name some living (biotic) and non-living (abiotic) parts of an ecosystem.	I can define key terms like community, population, habitat and explain how energy moves through a food web	I can explain the role of decomposers and interpret pyramids of numbers to show energy transfer in ecosystems.
Interactions within Ecosystems	I can identify some habitats found in the Sonoran Desert and other ecosystems.	I can describe interactions between organisms and their environment in different ecosystems.	I can explain the differences between habitats and ecosystems and how organisms are adapted to survive in these settings.
Adaptations and Competition	I can give examples of how plants or animals are adapted to their environments.	I can describe how adaptations help species survive and explain competition between organisms.	I can analyse mutualistic relationships and present research on how different species interact in ecosystems.
Invasive Species	I can name an invasive species and describe what a native species is.	I can describe how invasive species affect native plants and animals in an ecosystem.	I can explain why controlling invasive species is important and evaluate different methods of control.
Case Study - New Zealand (Debate)	I can identify an invasive species problem in New Zealand.	I can describe the ecological impact of invasive species and outline arguments for control.	I can construct and defend an argument in a debate using evidence about species control and conservation.
Bioaccumulation	I can state what an insecticide is and give a simple example.	I can describe how chemicals like DDT build up in organisms and impact the food chain.	I can explain why organisms at the top of the food chain have higher toxin levels and the environmental consequences.
Weather and Climate	I can describe the difference between weather and climate.	I can explain how weather affects plants and animals in an ecosystem.	I can analyse weather data to predict effects on local ecosystems and seasonal changes.
Climate and Atmosphere	I can name key gases in the atmosphere.	I can explain how greenhouse gases affect Earth's climate and describe human impacts.	I can evaluate how climate change may affect ecosystems and propose possible solutions.

What is a pesticide?

Pesticide
农药
농약

What is a pesticide?

Pests are animals/ insects/ plants that eat or harm plants that humans want to grow

Pesticide are chemicals that kill the pests

Pest
害虫
해충

Why do we use insecticides?

Insecticides are a type of pesticide/ chemical that kill insects.

There are 2 main reasons humans use insecticides…?



What is an insecticide?

1) Insecticides are very useful to farmers because it stops insects from eating their crops

Uses of insecticides?

2) To kill insects that cause disease, eg ………….

… malaria, dengue, bed bugs, lice/fleas



DDT

- **DDT is a type of insecticide**, it was first produced in the 1940s to kill mosquitos and fleas that were both spreading diseases
- It is very good at killing insects and was sprayed over large areas
- At the time, people thought it was only harmful to insects
- In 1962 people begun to realise that it was also harming the birds … but how?

DDT

There were 3 main reasons:

DDT

1) We now know that DDT does not break down over time and that it is a **persistent** chemical, meaning that it stays in the environment for many years.

   It also stays in the organisms that eat it.

When it is sprayed, some of it is carried into the air where it can be blown long distances.

Persistent

执着的

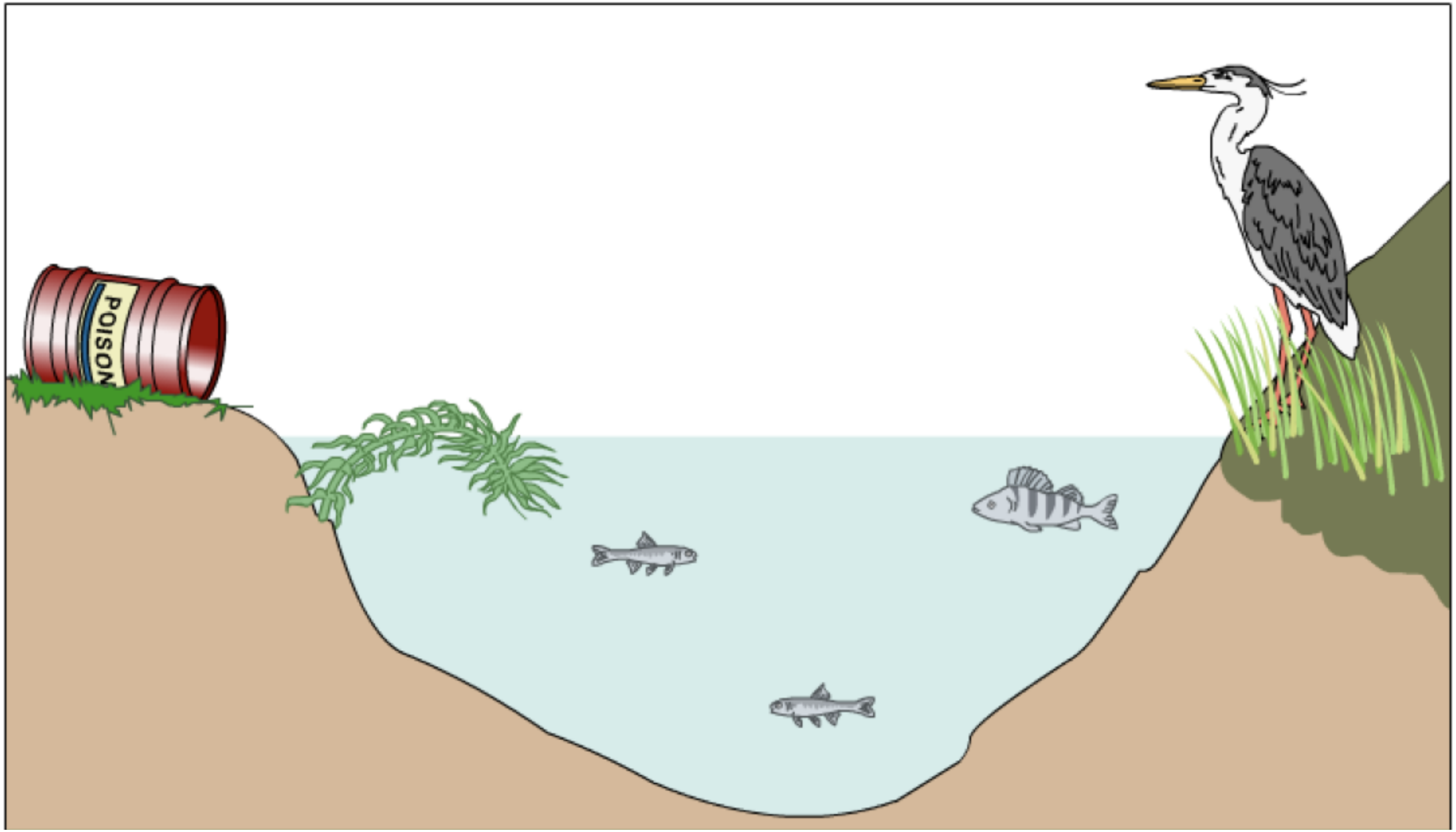
지속성 있는

DDT - How is it so harmful?

When DDT gets into an animals body it stays there for the whole life of the organism and never breaks down.

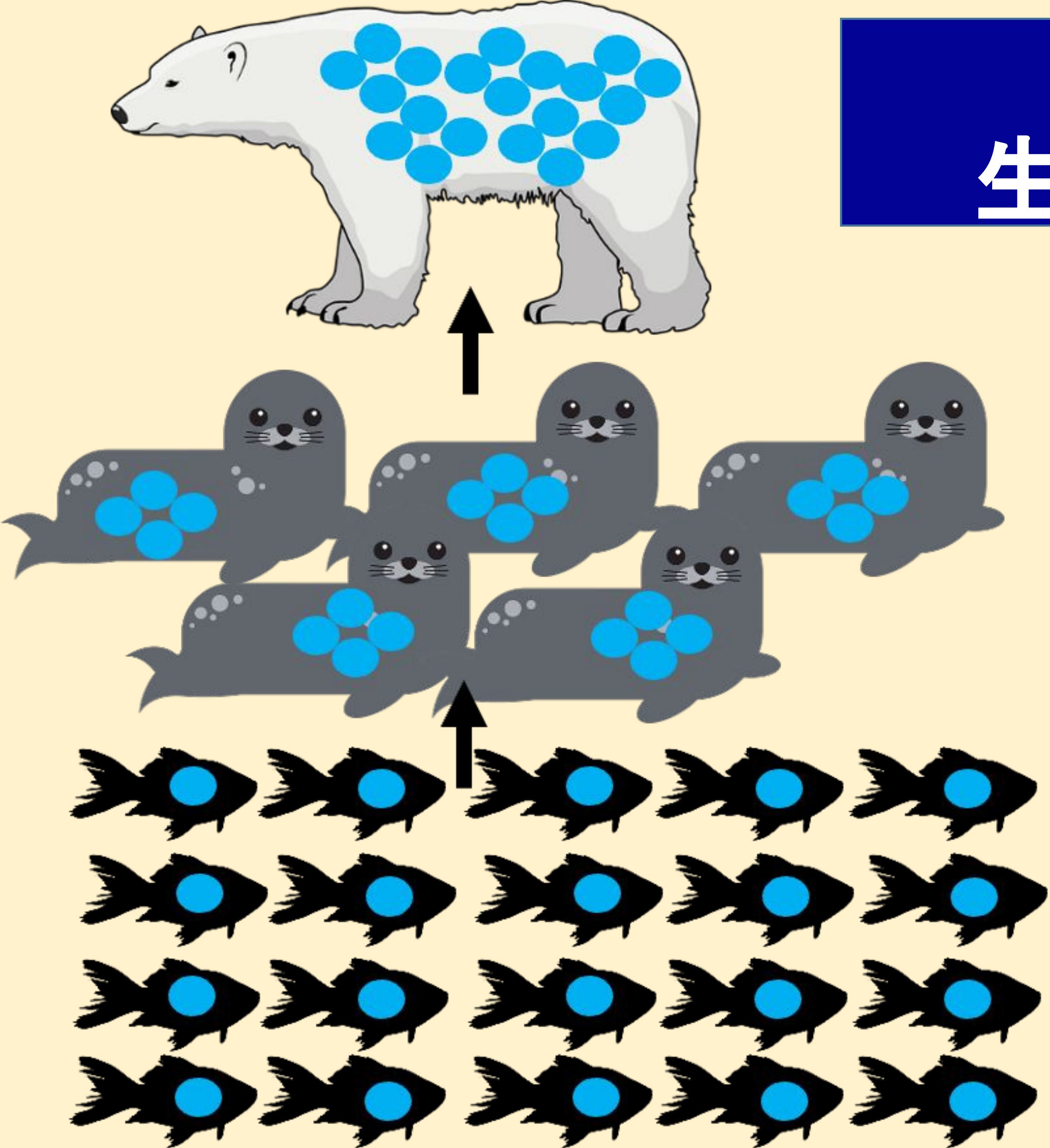
2) DDT is a **toxic substance** that is very harmful to many animals - making them ill or stopping them from being able to reproduce

Peregrine falcons nearly went extinct because of DDT



3) Why does the heron get more poison than the fish? What would happen to a predator that ate the heron?

Biomagnification
生物放大 / 생체 확대

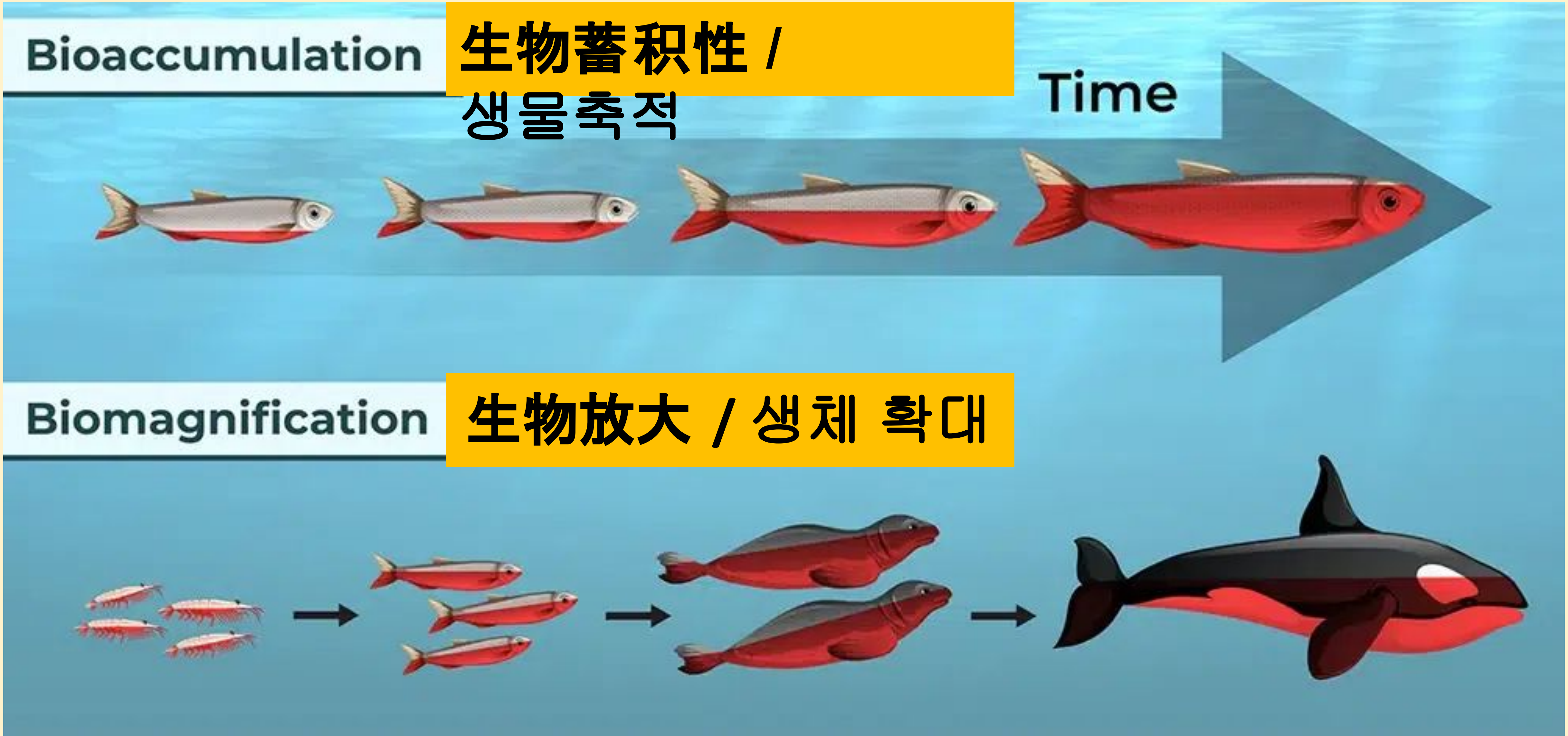


- If toxic substances are consumed or get into an animal's body it is passed along the food chain.
- The blue dots represents a **toxin**, which is a harmful chemical.

This is biomagnification

The gradual **build up** of a harmful toxin in an organism through a **food chain**.

What is the difference between biomagnification and bioaccumulation?



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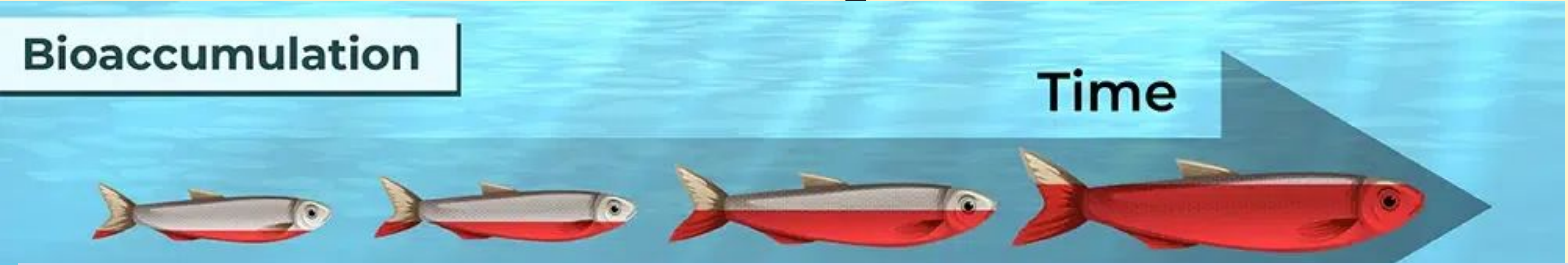
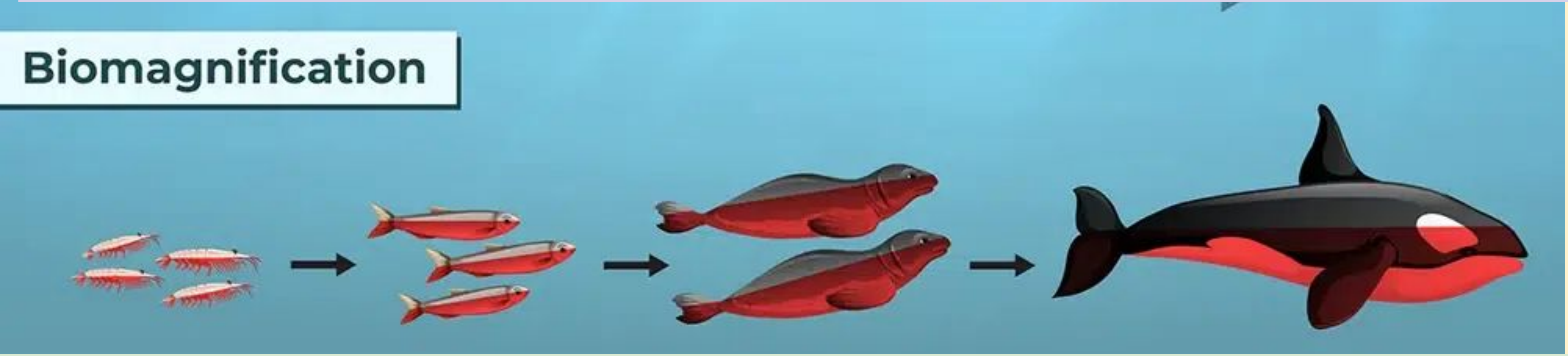


Bioaccumulation is the gradual build-up of toxins in an organisms



The gradual **build up** of a harmful toxin in an organism through a **food chain**.

What is the difference between biomagnification and bioaccumulation?

Aspect	Bioaccumulation	Biomagnification
Where it occurs		
How it happens		
Requires movement up food chain?		
Requires organism to be eaten?		
Key idea		

What is the difference between biomagnification and bioaccumulation?

Aspect	Bioaccumulation	Biomagnification
Where it occurs	In a *single organism* over its lifetime	Across *multiple organisms* in a food chain
How it happens		
Requires movement up food chain?		
Requires organism to be eaten?		
Key idea		

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Where it occurs	In a *single organism* over its lifetime	Across *multiple organisms* in a food chain
How it happens	Chemical builds up in one organism faster than it can be removed	Chemical concentration increases as it moves *up trophic levels*
Requires movement up food chain?		
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Key idea	Build-up within an individual	Build-up across a food chain

So pesticides like DDT

Are very damaging to the organisms for 3 main reasons:

1) They are **toxic 有毒的 / 독성**

2) They are **persistent 执着的 / 지속성 있는**

3) They **bioaccumulate** in an organism, causing **biomagnification** up the food chain.

DDT is a persistent poison. It builds up along the food chain.

DDT saves lives! It kills mosquitoes that carry malaria.

DDT helps protect crops from locusts. They can eat tonnes of food in a few days. DDT prevents people from starving to death.

We all have some DDT inside us now because it persists in the environment. We have poisoned ourselves with this dreadful chemical.

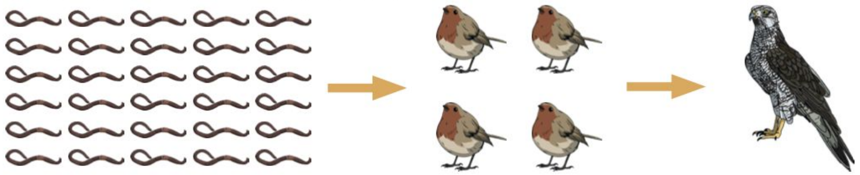
DDT has driven some birds of prey almost to extinction. It would have been better if we'd never invented it!

DDT - friend or foe? What do you think? And why?

Task: Complete the worksheet

Bioaccumulation

A simple food chain is represented below.



worm robin goshawk

Insecticides are chemicals which are sprayed on crops to kill insects. The chemicals also get into the soil where the worms live. Insecticides are chemicals that can undergo **bioaccumulation**.

1. Explain what is meant by the term bioaccumulation.

____

____

2. Explain how bioaccumulation could affect the goshawk.

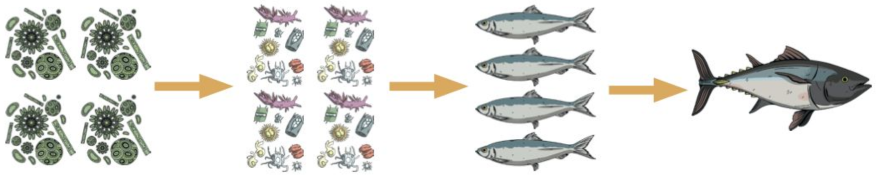
____

____

____

____

An ocean food chain is shown below.



phytoplankton zooplankton sardine tuna

Chemicals containing mercury used to be present in pesticides. Some of this mercury remains in the ocean and is absorbed by organisms that live there. Mercury damages the nervous and reproductive systems of mammals.

3. Explain why people are advised to limit the amount of tuna they eat during pregnancy.

____

____

____

____

1. Explain what is meant by the term bioaccumulation.

   Bioaccumulation is the gradual build-up/accumulation of toxins/chemicals in an organism.

2. Explain how bioaccumulation could affect the goshawk.

   The toxins/chemicals that accumulate in an organism are passed onto the next organism in the food chain via feeding/when they are eaten.
   Each robin eats many worms, and each goshawk eats four/multiple robins.
   This means that the goshawk will absorb greater amounts of the toxins/chemicals.
   These toxins may kill the goshawk or prevent the goshawk from reproducing.

Mark with a green pen!

3. Explain why people are advised to limit the amount of tuna they eat during pregnancy.
   Mercury will accumulate/build-up in the organisms in the food chain.
   The highest levels of mercury will accumulate in the tuna because it eats many fish (which have eaten many zooplankton, which have eaten many phytoplankton).
   Eating too much tuna during pregnancy could provide dangerous levels of mercury.
   The mercury could damage the nerves or reproductive system of the baby.

Mark with a green pen!

Microplastics 微塑料 / 미세플라스틱

Why do you think microplastics are bad for the environment and for the human population?

Exercise 4.4A

Pg 82-84 in the workbook

4.4 Bioaccumulation

Exercise 4.4A Microplastics

In this exercise, you will learn about microplastics. You can try finding information in a bar chart. You will also think about bioaccumulation.

Focus

Microplastics are tiny pieces of plastic less than 5 mm long.

Some microplastics come from big pieces of plastic that people have thrown away, and that slowly break apart. Some microplastics come from healthcare and beauty products such as toothpastes and face creams.

In the sea, microplastics can float in the water. They are accidentally taken into the bodies of living organisms when they feed. Some of the denser microplastics slowly fall to the sea bed.

1 What are microplastics?

....

....

2 Explain where microplastics come from.

....

....

....

3 Explain why some microplastics are found in the mud at the bottom of the sea.

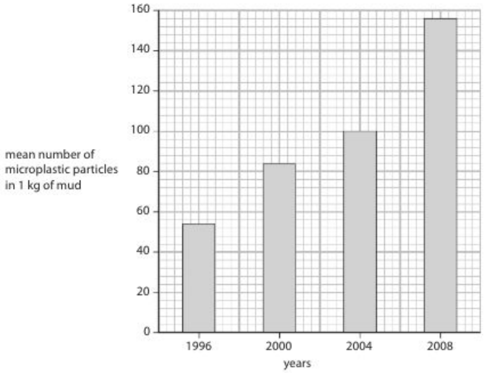
....

....

....

Practice

The graph shows the quantities of microplastics found in the mud at the bottom of the sea in four years between 1996 and 2008.



4 Look at the graph.

a State the mean number of microplastic particles found in 1 kg of mud in 1996.

....

b Calculate the increase in the number of microplastic particles in 1 kg of mud between 1996 and 2008.

Show your working.

....

Topic 4.4 Bioaccumulation

Exercise 4.4A Microplastics

1 tiny pieces of plastic less than 5 mm long

2 Some come from large pieces of plastic that break up into small pieces. Others are manufactured as microplastics, used in products such as face creams and toothpaste.

3 Some sink to the bottom because they are denser than water. Some go into the bodies of animals and may be carried to the bottom when the animal dies.

4 a 54

b $156 - 54 = 102$

5 As they feed, they take in microplastics that are floating in the water. There may also be some microplastics in the bodies of the zooplankton that they eat.

6 Bioaccumulation is the build-up of substances in an organism's body over its lifetime, because the substance does not break down in its body.

7 Seals eat many fish in their lifetimes, and all of the microplastics in the fish that they eat gradually build up in their bodies.

Mark with a green pen!